

Expt No: 04

LINE FOLLOWING ROBOT (2WD / 4WD)

Date: _____ / _____ / 2025

1. AIM:

To design and develop a Line Following Robot using Arduino UNO that detects and follows a path using IR sensors. The robot will be implemented for both 2-wheel drive (2WD) and 4-wheel drive (4WD) configurations.

2. APPARATUS REQUIRED:

- a. Microcontroller: Arduino UNO R3,
- b. Sensors: IR Sensor Array / IR Reflectance Sensors,
- c. Motor Driver: L293D / L298N Module,
- d. Chassis: 2WD or 4WD Robot Platform,
- e. Motors: DC Geared Motors (2 or 4 as per setup),
- f. Power Supply: 9V / 12V Battery,
- g. Miscellaneous: Jumper wires, Breadboard, USB Cable, Black Line Track (Black tape on white surface).

3. THEORY:

A Line Following Robot operates based on the reflectivity difference between black and white surfaces. Infrared sensors emit IR light and detect reflected light intensity. When placed over a white surface, the reflection is higher, whereas over a black line, reflection is minimal. The Arduino reads the digital output from these sensors to control motor direction—adjusting wheel speeds to ensure the robot follows the path.

4. PROCEDURE:

a) 2-Wheel Drive (2WD):

- i. Connect two IR sensors to Arduino digital pins 2 and 3.
- ii. Connect left and right motor pins to motor driver outputs.
- iii. Connect motor driver inputs to Arduino pins 8, 9, 10, and 11.
- iv. Upload the Line Follower code to the Arduino UNO.
- v. Place the robot on the black line path and observe movement.

--- Arduino Code (2WD) ---

```
const int leftSensor = 2;
const int rightSensor = 3;
const int leftMotor1 = 8;
const int leftMotor2 = 9;
const int rightMotor1 = 10;
const int rightMotor2 = 11;

void setup() {
  pinMode(leftSensor, INPUT);
  pinMode(rightSensor, INPUT);
  pinMode(leftMotor1, OUTPUT);
  pinMode(leftMotor2, OUTPUT);
  pinMode(rightMotor1, OUTPUT);
  pinMode(rightMotor2, OUTPUT);
}
```

```

void loop() {
  int leftState = digitalRead(leftSensor);
  int rightState = digitalRead(rightSensor);

  if(leftState == LOW && rightState == LOW) {
    // Move Forward
    digitalWrite(leftMotor1, HIGH);
    digitalWrite(leftMotor2, LOW);
    digitalWrite(rightMotor1, HIGH);
    digitalWrite(rightMotor2, LOW);
  }
  else if(leftState == LOW && rightState == HIGH) {
    // Turn Left
    digitalWrite(leftMotor1, LOW);
    digitalWrite(leftMotor2, LOW);
    digitalWrite(rightMotor1, HIGH);
    digitalWrite(rightMotor2, LOW);
  }
  else if(leftState == HIGH && rightState == LOW) {
    // Turn Right
    digitalWrite(leftMotor1, HIGH);
    digitalWrite(leftMotor2, LOW);
    digitalWrite(rightMotor1, LOW);
    digitalWrite(rightMotor2, LOW);
  }
  else {
    // Stop
    digitalWrite(leftMotor1, LOW);
    digitalWrite(leftMotor2, LOW);
    digitalWrite(rightMotor1, LOW);
    digitalWrite(rightMotor2, LOW);
  }
}

```

b) 4-Wheel Drive (4WD):

- i. Use four DC motors with a 4WD chassis.
- ii. Connect the motor driver outputs to all four motors (two per side).
- iii. Connect IR sensor array with at least 3 sensors for better precision.
- iv. Upload the modified 4WD code and place on the line path.
- v. Observe smoother turning and better line tracking.

--- Arduino Code (4WD) ---

```

const int leftSensor = 2;
const int middleSensor = 3;
const int rightSensor = 4;
const int leftMotor1 = 8;
const int leftMotor2 = 9;
const int rightMotor1 = 10;
const int rightMotor2 = 11;

void setup() {
  pinMode(leftSensor, INPUT);
  pinMode(middleSensor, INPUT);
  pinMode(rightSensor, INPUT);
  pinMode(leftMotor1, OUTPUT);
  pinMode(leftMotor2, OUTPUT);
  pinMode(rightMotor1, OUTPUT);
  pinMode(rightMotor2, OUTPUT);
}

void loop() {
  int left = digitalRead(leftSensor);
  int middle = digitalRead(middleSensor);
  int right = digitalRead(rightSensor);

  if(middle == LOW) {
    // Move Forward
    digitalWrite(leftMotor1, HIGH);
    digitalWrite(leftMotor2, LOW);
    digitalWrite(rightMotor1, HIGH);
    digitalWrite(rightMotor2, LOW);
  }
  else if(left == LOW) {
    // Turn Left

```

```
digitalWrite(leftMotor1, LOW);
digitalWrite(leftMotor2, LOW);
digitalWrite(rightMotor1, HIGH);
digitalWrite(rightMotor2, LOW);
}
else if(right == LOW) {
// Turn Right
digitalWrite(leftMotor1, HIGH);
digitalWrite(leftMotor2, LOW);
digitalWrite(rightMotor1, LOW);
digitalWrite(rightMotor2, LOW);
}
else {
// Stop
digitalWrite(leftMotor1, LOW);
digitalWrite(leftMotor2, LOW);
digitalWrite(rightMotor1, LOW);
digitalWrite(rightMotor2, LOW);
}
}
```

5. OBSERVATION:

The robot successfully followed the black line track. In 2WD mode, it performed basic tracking, while the 4WD configuration exhibited better stability and smoother cornering.

6. RESULT:

A Line Following Robot was successfully constructed and programmed using Arduino UNO. Both 2WD and 4WD configurations effectively followed the line track based on IR sensor feedback.